Maclaurin Series - Part 2

Questions 1–46

Question 1. Find $f'(x)$, $f''(x)$, $f^{(3)}(x)$ and $f^{(4)}(x)$ for each of the following:

a) $f(x) = \sin x$
b) $f(x) = \cos x$
c) $f(x) = e^x$
d) $f(x) = \ln(1+x)$
e) $f(x) = (1+x)^n$ for constant n

Question 2. For each of the following, derive the Maclaurin series by using up to and including $f^{(4)}(x)$:

a) $f(x) = \sin x$
b) $f(x) = \cos x$
c) $f(x) = e^x$
d) $f(x) = \ln(1+x)$
e) $f(x) = (1+x)^n$ for constant n

Question 3. By considering differentials, explain why the Maclaurin series for $\ln x$ does not exist.

Question 4. For each of the following, use differentiation to find the first three non-zero terms in the Maclaurin series and state the range of values for which the expansion is valid:

a) e^{5x}
b) $\sin 3x$
c) $\cos \frac{x}{2}$
d) $\ln(1+4x)$
e) $\ln(x+2)$
f) $(1+2x)^{1/4}$
g) $\dfrac{1}{(1-2x)^3}$

Question 5. Derive the first three non-zero terms of the Maclaurin series for:

a) $\cosh x$

b) $\sinh x$

Question 6. It is given that $f(x) = \sqrt{1+4x}$.

a) Find $f'(x)$ and $f''(x)$.

b) Determine the first three terms of the Maclaurin series for $\sqrt{1+4x}$.

c) State the validity of the Maclaurin series for $\sqrt{1+4x}$.

d) Hence, using a suitable value for x, show that $\sqrt{6} \approx \frac{39}{16}$.

Question 7.

a) Use differentiation to find the first three non-zero terms in the Maclaurin series for $\ln\left(\dfrac{x+2}{x+1}\right)$.

b) Determine the values for which the expansion is valid.

Question 8. It is given that

$$\sin\left(x-\frac{\pi}{4}\right)=\frac{\sqrt{2}}{2}\left(-1+x+\frac{1}{2}x^2-\frac{1}{6}x^3-\frac{1}{24}x^4+\cdots\right).$$

a) Use the series expansions of $\sin x$ and $\cos x$ to prove the result.

b) Use differentiation to prove the result.

Question 9. It is given that

$$\cos\left(x+\frac{3\pi}{4}\right)=\frac{\sqrt{2}}{2}\left(-1-x+\frac{1}{2}x^2+\frac{1}{6}x^3-\frac{1}{24}x^4+\cdots\right).$$

a) Use the series expansions of $\sin x$ and $\cos x$ to prove the result.
b) Use differentiation to prove the result.

Question 10.

a) Using differentiation, find the Maclaurin series of $\sin 2x$ up to and including the x^3 term.
b) Using differentiation, find the Maclaurin series of $\cos 2x$ up to and including the x^4 term.
c) Hence, given x is small such that terms in x^5 and higher powers are negligible, show that

$$\cos\left(2x+\frac{\pi}{6}\right)\approx\frac{\sqrt{3}}{2}-x-\sqrt{3}x^2+\frac{2x^3}{3}+\frac{\sqrt{3}x^4}{3}.$$

Question 11.

a) Use differentiation to find the Maclaurin series for $\sin\left(\frac{\pi}{3}+x\right)$ up to and including the x^2 term.

b) Hence show that when h is small,

$$\int_0^h x^2 \sin\left(\frac{\pi}{3}+x\right) dx \approx \frac{\sqrt{3}h^3}{6} + \frac{h^4}{8} - \frac{\sqrt{3}h^5}{20}.$$

Question 12. Use the Maclaurin series for $\cos 2x$ to find the series expansion for $\cos^2 x$ up to and including the x^6 term.

Question 13.

a) Use the Maclaurin series expansion for $\cos x$ to find the series expansion of $\sin^2(\frac{x}{2})$ up to and including the x^4 term.

b) Hence find an approximation to $\displaystyle\int_{\pi/3}^{\pi/2} \left(\frac{1}{x}\sin^2(\frac{x}{2})\right) dx$, giving your answer to two decimal places.

Question 14.

a) Use the Maclaurin series for $\cos 2x$ to find the series expansion for $\sin^2 x$ up to and including the x^6 term.

b) Hence, using your result from part (a), find an approximation for $\displaystyle\int_0^{\pi/2} \sin^2 x\, dx$.

c) Evaluate $\displaystyle\int_0^{\pi/2} \sin^2 x\, dx$.

d) Find the percentage error in your approximation, giving your answer to two significant figures.

Question 15. It is given that $f(x) = \ln(1 - x + \sin x)$ and that the constant k lies in the domain of f.

a) Show that k can never be greater than 2.

b) Find $f'(x)$.

c) By using $\cos x \approx 1 - \frac{1}{2}x^2$ and $\sin x \approx x$, determine a cubic approximation for $f(x)$ for small values of x.

Question 16. It is given that $f(x) = \cosh^2 x$.

a) Show that $f''(x) = 2\cosh(2x)$.

b) Explain why the coefficients of the odd powers of x in the Maclaurin series for $\cosh^2 x$ are all zero.

c) Given n is a positive even number, find the coefficient of x^n in this series.

Question 17. It is given that $f(x) = \sinh^n x$ for $n \geq 2$.

a) By using differentiation find the first three non-zero terms of the Maclaurin series of $\sinh(x)$.

b) Hence, find the first three non-zero terms of the Maclaurin series of $\sinh^n(x)$.

Question 18.

a) Use differentiation to find the first two non-zero terms of the Maclaurin expansion of $\ln(1 + \cos x)$.

b) By considering the root of the equation $\ln(1 + \cos x) = 0$, deduce that $\pi \approx 2\sqrt{2}$.

Question 19. Use differentiation to find the first two non-zero terms of the Maclaurin series of $\sec x$.

Question 20.

a) Use differentiation to show that the first three non-zero terms of the Maclaurin series of $\cos x$ are $1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$.

b) Hence, given that x is small, find a and b if $\sec x = 1 + ax^2 + bx^4$.

Question 21.

a) Use differentiation to show that the first three non-zero terms of the Maclaurin series of $\sin x$ are $x - \frac{1}{6}x^3 + \frac{1}{120}x^5$.

b) Hence, given that x is small, find a and b if $\operatorname{cosec} x = \frac{1}{x} + ax + bx^3$.

Question 22. It is known that $f(x) = \sin(x^3)$.

a) Find $f'(x)$.

b) Use a series expansion to find an approximation to $f'(x)$ in terms up to and including the x^{20} term.

c) Integrate your answer in part (b).

d) State the series expansion of $\sin(x^3)$ as far as the term in x^{21}.

Question 23. It is given that $f(x) = (1-x)^2 e^x$.

a) Show that $f''(x) = (x^2 + 2x - 1)e^x$.

b) Hence write $(1-x)^2 e^x$ in ascending powers of x up to and including the x^2 term.

Question 24. It is given that $f(x) = (1-x)\sin x$.

a) Show that $f''(x) = (x-1)\sin x - 2\cos x$.

b) Hence write $(1-x)\sin x$ in ascending powers of x up to and including the term in x^3.

Question 25. It is given that $f(x) = e^{2x} \sin x$.

a) Find $f(0)$ and $f'(0)$.

b) Show that $f''(x) = 4f'(x) - 5f(x)$.

c) Hence find the Maclaurin series for $f(x)$ up to and including the x^3 term.

Question 26. It is given that $f(x) = e^{-x} \cos x$.

a) Find $f(0)$ and $f'(0)$.

b) Show that $f''(x) = -2f'(x) - 2f(x)$.

c) Hence find the Maclaurin series for $f(x)$ up to and including the x^3 term.

Question 27. It is given that $f(x) = \sin x \cosh x$ and $f^{(4)}(x) = kf(x)$, where k is an integer.

a) Find the value of k.

b) Hence determine the first three non-zero terms of the Maclaurin series for $f(x)$.

Question 28. Determine the Maclaurin series of e^{x^2+1} up to and including the x^3 term:

a) Using differentiation.

b) Without using differentiation.

Question 29. It is given that $f(x) = \dfrac{1}{1+e^{-x}}$.

a) Show that $f''(x) = \dfrac{e^x(1-e^x)}{(e^x+1)^3}$.

b) Hence find the first three non-zero terms of the Maclaurin series of $\dfrac{1}{1+e^{-x}}$.

Question 30.

a) Show that $\dfrac{d^3}{dx^3}(\tan x) = 2\sec^2 x(\sec^2 x + 2\tan^2 x)$.

b) Use differentiation to find the Maclaurin series for $\tan x$ up to and including the term in x^3.

c) Hence show that when h is small, $\displaystyle\int_h^{3h} \frac{\tan x}{2x}\,dx \approx h + \frac{13}{9}h^3$.

Question 31. It is given that $f(x) = \ln(\cos x)$.

a) Show that $f'(x) = -\tan x$.

b) Show that $\dfrac{d}{dx}(\sec^2 x) = 2\tan x \sec^2 x$.

c) Write $\ln(\cos x)$ as a Maclaurin series in ascending powers of x up to and including the x^4 term.

d) Hence find an approximation for $\ln(\frac{\sqrt{3}}{2})$.

e) Find the percentage error in the approximation, giving your answer to three significant figures.

Question 32.

a) Show that $\dfrac{d}{dx}(\ln(\sin x)) = \cot x$.

b) Hence, by using the first three terms of the Maclaurin series for $\sin x$ and $\ln(1+x)$, find the first three non-zero terms in the expansion of $\cot x$.

Question 33.

a) Show that $\frac{d}{dx}(\ln(\tan(\frac{x}{2}))) = \operatorname{cosec} x$.

b) Given that $\tan x \approx x + \frac{x^3}{3} + \frac{2x^5}{15}$, find the first three non-zero terms in the expansion of $\operatorname{cosec} x$.

Question 34. It is given that $f(x) = \arcsin x$.

a) Show that $f'(x) = \frac{1}{\sqrt{1-x^2}}$.

b) Hence find the first two non-zero terms of the Maclaurin series for $\arcsin x$.

Question 35. It is given that $f(x) = \arccos x$.

a) Show that $f'(x) = -\dfrac{1}{\sqrt{1-x^2}}$.

b) Hence find the first three terms of the Maclaurin series for $\arccos x$.

Question 36. It is given that $f(x) = \arctan x$.

a) Show that $f'(x) = \dfrac{1}{1+x^2}$.

b) Hence find the first two non-zero terms in the Maclaurin expansion of $\arctan x$.

c) Show that $\dfrac{\pi}{4} = 1 - \dfrac{1}{3} + \cdots$.

Question 37.

a) Find the first four non-zero terms of the Maclaurin series for $y = \dfrac{1}{\sqrt{1+x^2}}$.

b) Hence find the first four non-zero terms of the Maclaurin series for $y = \operatorname{arsinh} x$.

Question 38. Show that the first four non-zero terms of the Maclaurin series for $\operatorname{artanh} x$ are given by $x + \dfrac{x^3}{3} + \dfrac{x^5}{5} + \dfrac{x^7}{7}$.

Question 39.

a) Find the first three terms of the Maclaurin series for $\arccos x$.

b) Show that the Maclaurin series expansion for $(\arccos x)\ln(1+x)$ up to and including the x^3 term is $\frac{\pi}{2}x+(-1-\frac{\pi}{4})x^2+\frac{1}{6}(3+\pi)x^3$.

Question 40.

a) Use differentiation to find the Maclaurin series for $\arctan(1+x)$ up to and including the x^2 term.

b) Hence show that when h is small, $\int_{-h}^{h} x\arctan(1+x)\,dx \approx \frac{1}{3}h^3$.

Question 41.

a) Find the first two terms of the Maclaurin series for $\arcsin x$.

b) Hence find an approximation to $\int_0^{1/2} \arcsin x \, dx$, giving your answer to three significant figures.

c) Determine an exact value of $\int_0^{1/2} \arcsin x \, dx$.

d) Find the percentage error in your approximation, giving your answer to one decimal place.

Question 42. It is given that $f(x) = \arcsin(x + \frac{1}{3})$.

a) Find constants a and b if the Maclaurin expansion of $f(x)$ is $\arcsin(\frac{1}{3}) + ax + bx^2 + \cdots$.

b) Hence find an approximate value of $\int_0^{0.05} f(x) \, dx$, giving your answer to three significant figures.

Question 43. It is given that $f(x) = \arccos(2x)$.

a) Find $f'(x)$.

b) By using a standard series to expand $f'(x)$, find the series for $f(x)$ up to and including the term in x^5.

Question 44.

a) Derive the Maclaurin series of $\arcsin(2x)$ up to and including the x^3 term.

b) By substituting $x = \frac{1}{4}$, show that $\pi \approx 6(\frac{1}{2} + \frac{1}{48})$.

c) Find the percentage error in your answer to part (b).

Question 45. It is given that $f(x) = \text{artanh}(\frac{1-x}{3+x})$ for $x > -1$.

a) Show that $f'(x) = -\dfrac{1}{2x+2}$.

b) Find $f''(x)$ in its simplest form.

c) Hence show that the first three non-zero terms of the Maclaurin series for $f(x)$ are $\dfrac{1}{2}\ln 2 - \dfrac{1}{2}x + \dfrac{1}{4}x^2$.

Question 46. It is given that $f(x) = \arctan(\cosh x)$.

a) Show that $f''(x) = \dfrac{\cosh x(1 + \cosh^2 x - 2\sinh^2 x)}{(1+\cosh^2 x)^2}$.

b) Find the Maclaurin series for $f(x)$ up to and including the x^2 term.